ECE750T: Research Project Proposal

Reasoning Under Observation: Observer Effects on LLM Rationales

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Q1: Research Question

How do observer identity and the declared visibility of a language model's verbalized reasoning affect its accuracy, susceptibility to an anonymous suggested answer, and disclosure of that influence? I will compare a developer, a domain expert, and an automated AI monitor when each is said to see either the model's rationale and answer or only its answer. This connects evaluation, auditing, and foundation-model safety: if announced oversight changes behaviour, an audit may not predict deployment.

Q2: Prior Work and Limitations

Covas and Hidalgo Toledo [2026] report that human and AI monitoring prompts change style, but study one debate topic rather than correctness or disclosure. Feng et al. [2026] find that reasoning can reduce sycophancy while masking it through post-hoc rationalization, but do not vary who observes it. Li et al. [2026] separate evaluation cues, recognition, and behavioural response; recognition often causes no response. Chen et al. [2026] find that models control outputs more readily than reasoning traces. Turpin et al. [2023] show that suggested answers can influence predictions without acknowledgment. I combine these concerns in one audit.

Q3: Novelty of Approach

Prior work separately studies observer identity, evaluation awareness, sycophancy, and reasoning controllability. I will cross observer identity with declared rationale visibility while holding an anonymous suggestion fixed. Unlike style-based observer studies, I measure correctness and disclosure; unlike model-organism studies, I require no training or deceptive objective.

Q4: Impact and Motivation

Chain-of-thought is proposed as audit evidence. If monitoring changes a rationale without comparably changing the answer—or encourages silent conformity—auditors may overestimate transparency. The results could guide benchmark designers deciding when verbalized reasoning is useful evidence.

Q5: Risks and Assumptions

Generated rationales are not guaranteed to reveal internal computation, so I will not claim intentional concealment. "Declared-private" describes the prompt, not actual privacy or belief. Observer wording may confound identity with authority; matched templates and a second paraphrase will test this. A pilot will avoid floor/ceiling tasks; blinded human coding will check automated labels. Benchmark exposure and one-model generalization remain limitations.

Q6: Resources

I will use 100–150 public BIG-Bench Hard multiple-choice items [Suzgun et al., 2023] from three tasks selected by a pilot, plus a random anonymous answer suggestion per item. The primary model will be Qwen3-4B in Q4_K_M quantization, run through `llama.cpp` at 4K–8K context and batch size one. Its 2.5 GB weights and approximately 4 GB runtime footprint fit my 8 GB RTX 3070 Ti. Python will generate prompts, parse answers, and score results; no training is required.

Q7: Weekly Plan

Week	Plan
Oct 12–18	Finalize literature, prompt templates, annotation rubric, and analysis protocol.
Oct 19–25	Implement inference and scoring pipeline; acquire BBH data and quantized model.
Oct 26–Nov 1	Pilot tasks and prompts; choose three tasks and freeze primary conditions.
Nov 2–8	Run the main eight-condition experiment and check output parsing/data quality.
Nov 9–15	Analyze accuracy, hint-following, answer flips, and paired uncertainty estimates.
Nov 16–22	Code rationales automatically; blind-label a stratified subset and validate labels.
Nov 23–29	Run paraphrase/seed robustness checks; draft report figures, results, and limitations.
Nov 30–Dec 4	Complete and revise the report; prepare the project presentation and artifacts.

Q8: Milestones and Success Criteria

The eight conditions are three observers × visible/declared-private rationale, a no observer/no suggestion baseline, and an anonymous hint-only control. Suggested labels are random; analyses of influence focus on items where the suggestion is wrong. Primary outcomes are accuracy, following the wrong suggestion, and "silent conformity" (following it without acknowledgment). Secondary outcomes include answer flips, uncertainty language, observer mentions, and rationale–answer consistency. Answers will be scored deterministically. Rationale categories will be assigned automatically or by an LLM judge, then checked against blinded human labels on a stratified subset. I will report paired bootstrap 95% confidence intervals and McNemar tests for binary answer changes. Success means a reproducible pipeline, a documented manipulation check, validated rationale coding, and bounded effect estimates; a precise null result is informative because it limits observer effects in this setting.

Tools and Assistance

I used Cursor-supported language models for literature discovery, research planning, and proposal drafting. I expect to use them for coding assistance and possibly preliminary rationale labeling. I will verify citations, code, labels, and experimental results myself, document final tool use, and retain human-checked labels for judge validation.

References

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